



Course - DSE & DBD
course code - 25SC1302E

SignEase

Breaking barriers through sign language — a full-stack translator with real-time, in-browser hand recognition.

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THE PROBLEM

Communication shouldn't require a shared language.



Schools & Colleges

Deaf and hard-of-hearing students often lack real-time tools to communicate in mixed classrooms.



Workplaces

Everyday professional interactions default to spoken/written language, excluding sign-language users.



Public Spaces

Hospitals, transit, and government offices rarely offer any live translation aid at all.

WHAT SIGNEASE DOES

Everything you need to communicate



Real-Time Translation

Live camera feed with hand tracking.



Sign Dictionary

Searchable library of common signs.



User Accounts

Profiles, favorites, and feedback.

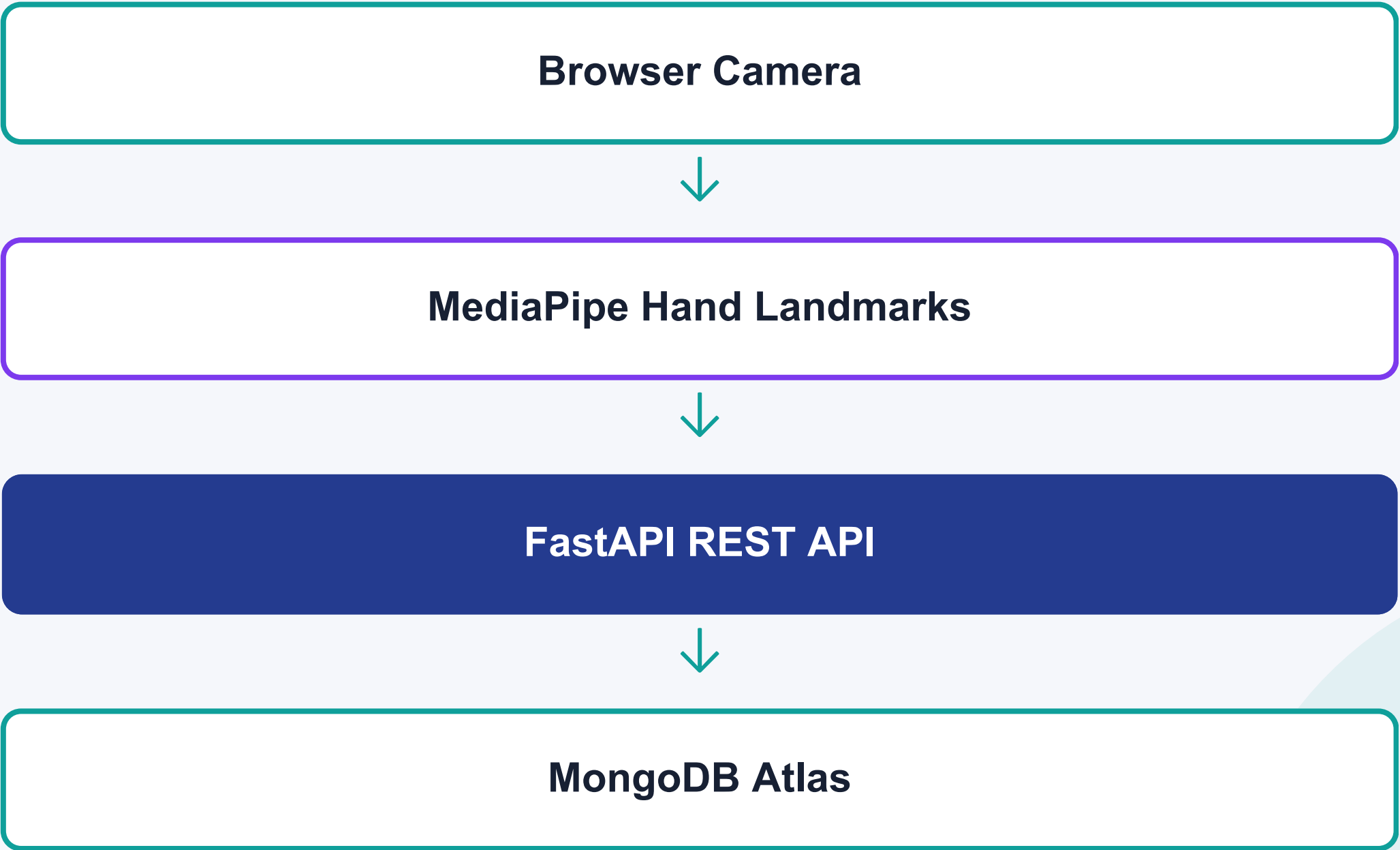


Learning Mode

Practice signs and track progress.

Tech stack & architecture

Frontend	HTML / CSS / JavaScript (SPA)
Backend	FastAPI (Python), JWT auth
Database	MongoDB (Atlas / Compass)
ML Recognition	MediaPipe Hands + k-NN, in-browser
Hosting	Render (backend) + static hosting (frontend)



From hand pose to predicted sign

1

Capture landmarks

MediaPipe Hands (pretrained) extracts 21 (x, y, z) points per hand, live in the browser.

2

Collect labeled samples

A built-in Train Signs tool records landmark snapshots labeled with a sign name.

3

Normalize

Each sample is translated to the wrist origin and scaled, making it position/size invariant.

4

Classify (k-NN)

Live landmarks are compared to saved samples; nearest neighbors vote on the predicted sign.

OVERVIEW

Six course outcomes, six implemented topics

CO	Topic chosen	Where it lives in SignEase
C01	Index Design	Automatic MongoDB indexes created on startup
C02	MongoDB Document Engineering (CRUD)	translations / favorites / users collections via Motor
C03	Authentication & Security (JWT)	/auth/login issuing bearer tokens, bcrypt password hashing
C04	Architectural Thinking (applied)	Domain-separated routes/services inside one FastAPI app
C05	API Gateway pattern (applied)	FastAPI as single entry point: routing, auth, error handling
C06	Documentation & Final Showcase	README, render.yaml/Procfile, this presentation

Future work

- 1 Collect 30-50+ samples per sign to improve classifier accuracy
- 2 Support motion-based signs using landmark sequences, not single frames
- 3 Two-hand recognition for signs that use both hands
- 4 Deploy backend to Render with MongoDB Atlas for public access